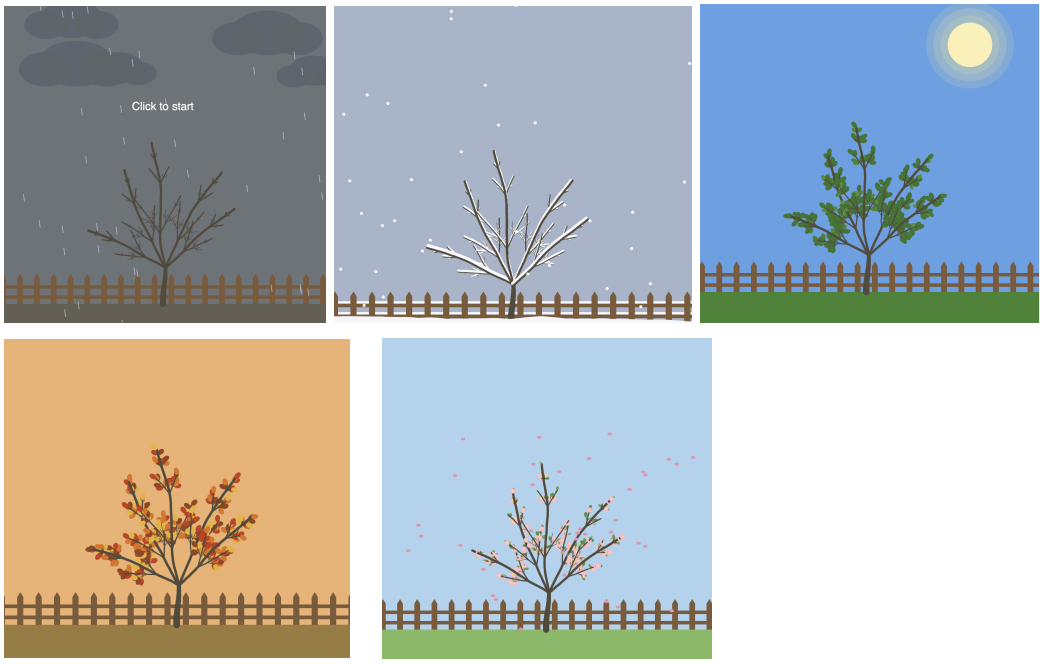
Ian Zimmerman Project 2 - Assignment: Trees

Brief Write Up:

Using an L-system, I designed a tree to mimic the maple trees I grew up with in Vermont. While coming up with this project, my mom sent me photos of foliage, which inspired me to tell the story of a tree as it progresses through the four seasons. Since high school, I have always studied while listening to classical music, and I came up with the idea to pair Vivaldi's *The Four Seasons* with the four seasons the tree experiences. In my scene, I tried to make the tree seem as if it were dancing to the music, telling a story with both sounds and images.

I took the basic L-system code we were given and modified it to make my tree mimic a maple tree. I then used a similar strategy from my first project and created a set of boolean variables for each season. One challenge I faced was making the scene react to the music. With the help of AI, I created timestamps and measured the music's volume so the code could react to changes in the music. The timestamp determines the season, and therefore when the leaves grow, change color, or fall, while the volume determines the intensity of the weather. In this way, using AI helped me learn some more advanced skills required to achieve my vision. Another challenge I faced was getting the snow to land on the branches. In real life, the way snow accumulates on branches depends on both the branch angle and how strong or thick it is. To recreate this in my code, I used sin() to measure the branch's flatness, along with a thickness variable, to determine how much snow each branch could hold.

Image(s):



Code: https://editor.p5js.org/Ian_Zimmerman/sketches/ZiT6xI-U_